MESHCORE SOLAR NODE ENCLOSURE

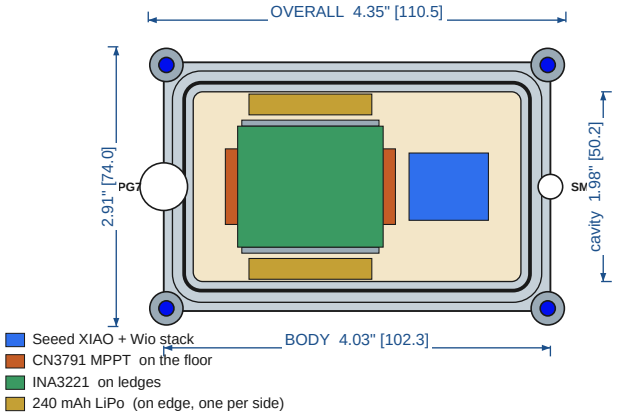
3-D printed weatherproof box · PETG / ASA · Build drawing · Sheet 1 of 1

SCALE 1 : 2 (views)
DIMENSIONS inches [mm]

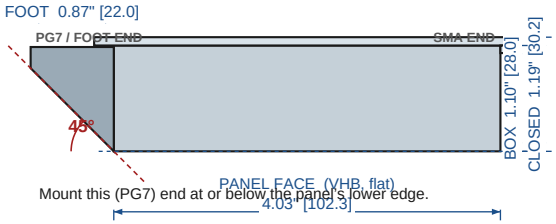
ISOMETRIC · EXPLODED



TOP · LID REMOVED (1 : 2)

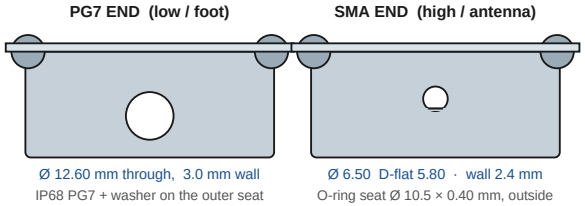


SIDE · CLOSED, FOOT ON (1 : 2)

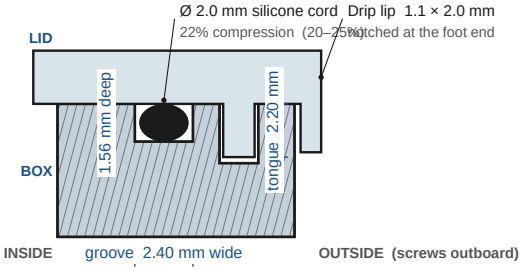


Mount this (PG7) end at or below the panel's lower edge.

ENDS (1 : 2)



SECTION A–A SEAL (3.2×)



OVERALL (CLOSED, FOOT ON)

Length (foot → SMA)
5.10" [129.5 mm]

Width (over the lid)
3.00" [76.2 mm]

Height (panel → lid)
1.19" [30.2 mm]

Box body
4.03" × 2.59" [102.3 × 65.8 mm]

Internal cavity
3.41" × 1.98" × 1.02" [86.7 × 50.2 × 26.0 mm]

Corner bosses and foot pads sit outside the body.
Battery pockets: edit bat_l, bat_w, bat_t.

SEALING SPECIFICATION

Primary seal	Ø 2.00 mm silicone cord (50–70 A), rectangular gland
Compression	22% design (range 20–25%). Groove depth 1.56 mm.
Groove	2.40 mm wide × 1.56 mm deep, 4.5 mm CL corner radius
Gland fill	84% when the lid is fully closed (target 70–90%)
Labyrinth	Tongue-and-groove INBOARD of the O-ring: 1.30 × 2.20 mm tongue, 0.15 mm radial clearance, 0.25 mm extra groove depth
Drip path	Rain lip 1.10 × 2.00 mm, 0.25 mm gap; notched at the foot end
Lid screws	4× M3 SHCS OUTSIDE the O-ring, into blind heat-set inserts
Inserts	M3 × 5.7 (Ruthex or equal), Ø 4.00 mm bore, 6.00 mm deep, BLIND
Screw length	M3 × 10 mm SS; optional M3 nylon washer under the cap
PG7 gland	Ø 12.60 mm hole, 3.00 mm wall, Ø 18 mm washer seat. IP68 PG7 with its NBR/EPDM gasket. Cable range 3–6.5 mm.
SMA bulkhead	Ø 6.50 mm D-hole, 5.80 mm flat toward the panel. Flat O-ring seat Ø 10.50 × 0.40 mm on the OUTSIDE. Wall 2.40 mm.
USB-C	NO opening.
Cord length	11.29" [287] cut, scarf-joined in the box groove

ASSUMED PARTS · PRINT

Battery ×2	502030-class 240 mAh 1S pouch WITH PCM. Assumed 32.5 × 20.2 × 5.5 mm (FLP-502030 is 32.5 × 20.2 × 5.1). Pocket is PARAMETRIC (bat_l, bat_w, bat_t) + swell. Cells stand ON EDGE, one slot on each side.
Seeed stack	XIAO nRF52840 21.0 × 17.8 mm on Wio-SX1262. Official stacked height 21 mm; pocket allows 24 mm.
INA3221	MusRock B0FTF9LC1T (MCU-3221), 38.3 × 31.7 mm PCB, 8 mm with parts. Sits on 45° ledges over the MPPT.
CN3791 MPPT	Common mini module 45 × 20 × 9.5 mm, on the floor.
Print	PETG or ASA. 0.20 mm layers, 0.40 mm nozzle, 4 walls, 5 top/bottom, 40% gyroid, 6 walls at insert bosses. Designed XY clearance 0.15 mm. Tight tolerances.
Orientation	BOX: panel face on the bed. LID: outer face on the bed (tongue up). FOOT: mating face on the bed (45° overhang). No supports.
Inserts	Iron in at the maker's temperature for the filament.
Foot screws	2× M3 × 10–12 mm into 4.5 mm blind inserts in the pads.

PROJECT
MeshCore solar monitor · Boardman / Hermiston, OR
Material: PETG or ASA | Wall 2.0 mm | No supports

FILES
`solar_node_enclosure.scad` (parametric source)
`solar_node_{box,lid,foot}.stl`

REV
A 12 Nov 2026 Initial release
Do not scale the drawing. Use written dimensions.